

The importance of the core groupoid of a category

Ross Street
School of Mathematical and Physical Sciences
CoACT, Macquarie University, Australia

From Darramurragal country

To Research Group on Representation Theory
and Homological Algebra, in China

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References

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- ▶ Teimuraz Pirashvili, Dold-Kan type theorem for Γ -groups, *Mathematische Annalen* **318** (2000) 277–298.
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- ▶ (with Stephen Lack) Combinatorial categorical equivalences of Dold-Kan type, *Journal of Pure and Applied Algebra* **219** (2015) 4343–4367.
- ▶ Nicholas J. Kuhn, Generic representation theory of finite fields in nondescribing characteristic, *Advances in Mathematics* **272** (2015) 598–610.
- ▶ The core groupoid can suffice, *Theory and Applications of Categories* **41(21)** (2024) 686–706.

Stop Press

- ▶ After preparing for this talk, the following article appeared on the math arXiv:
- ▶ Nate Harman, Andrew Snowden, and Elad Zelingher, Representations of finite matrix monoids, arXiv:2512.02226v1 (31 pages).
- ▶ The paper is very interesting in its use of programming (SageMath) and artificial intelligence (Claude.ai) to obtain an explicit formula for the idempotent given by Laci Kovács in 1992 without explicitly inverting a big invertible triangular matrix. The modern technology did not give the final result but did provide enough hint for the three authors to finish the job.

The general problem

- ▶ Let R be a commutative ring. Let $\mathcal{V} = \text{Ab}^R$ be the monoidal category of R -modules. For any category $\mathcal{F}$, we call the functor category $[\mathcal{F}, \mathcal{V}]$ the *category of R -linear representations of $\mathcal{F}$* .
- ▶ I am interested in categories $\mathcal{F}$ for which there is a groupoid $\mathcal{G}$ such that the categories $[\mathcal{F}, \mathcal{V}]$ and $[\mathcal{G}, \mathcal{V}]$ of representations are equivalent.
- ▶ In particular, $\mathcal{G}$ could be the core groupoid $\mathcal{F}_{\text{inv}}$ of $\mathcal{F}$; that is, the subcategory with the same objects and with only the invertible morphisms.
- ▶ We could freely split some idempotents in $\mathcal{F}$ without changing the representations. However, adding the new objects may change the core groupoid.

Linearizing

- ▶ Every category $\mathcal{F}$ gives rise to a $\mathcal{V}$ -category (that is, an R -linear category), denoted $R\mathcal{F}$, with the same objects and with hom R -module $R\mathcal{F}(A, B)$ free on the homset $\mathcal{F}(A, B)$.
- ▶ Indeed, $R\mathcal{F}$ is the free $\mathcal{V}$ -category on $\mathcal{F}$ so that the $\mathcal{V}$ -functor category $[R\mathcal{F}, \mathcal{V}]$ is isomorphic to the ordinary functor category $[\mathcal{F}, \mathcal{V}]$ with the pointwise R -linear structure.
- ▶ In these terms, we are interested in when $R\mathcal{F}$ and $R\mathcal{G}$ are **Morita equivalent $\mathcal{V}$ -categories** so that a groupoid suffices for all linear representations of $\mathcal{F}$.
- ▶ In Section 5 of Lack-St (2015) we showed that proving a Dold-Kan-type equivalence of categories reduced to solving a “core groupoid suffices” problem.

An example without details

- ▶ Indeed the original Dold-Kan equivalence between **simplicial R -modules** and **chain complexes of R -modules** reduces to the “core groupoid suffices” problem for $\mathcal{F} = \Delta_{\perp}$.
- ▶ Here $\Delta_{\perp}$ is the category of finite non-empty ordinals

$$\mathbf{a} = \{0, 1, \dots, a - 1\}, \quad a \in \mathbb{N}_{>0},$$

and first-element-and-order-preserving functions $\xi : \mathbf{a} \rightarrow \mathbf{b}$.

- ▶ As the only invertible morphisms in $\Delta_{\perp}$ are identities, the core groupoid in this case is discrete with countably many objects; so the category of representations is the product $\mathcal{V} \times \mathcal{V} \times \dots$ of countably many copies of $\mathcal{V}$, that is, the category of objects of $\mathcal{V}$ graded by the positive integers.

A baby example with details, page 1

- ▶ Let $C = \mathbf{3} = \{0, 1, 2\} \in \Delta_{\perp}$. The monoid $\Delta_{\perp}(C) = \Delta_{\perp}(\mathbf{3}, \mathbf{3})$ has six elements, namely, the functions

$$\iota = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \quad \sigma_0 = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \quad \sigma_1 = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\tau = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 2 & 2 \end{bmatrix} \quad v = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \quad \zeta = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

- ▶ A presentation of this monoid is provided by the generators σ_0 , σ_1 and τ subject to the relations that the generators are idempotents and

$$\sigma_0 \sigma_1 \sigma_0 = \sigma_0 \sigma_1 = \sigma_1 \sigma_0 \sigma_1$$

$$\tau \sigma_0 = \sigma_0, \tau \sigma_1 = \tau = \sigma_0 \tau, \sigma_1 \tau = \sigma_1.$$

A baby example with details, page 2

- ▶ The only invertible element in $\Delta_{\perp}(C)$ is the identity ι .
- ▶ The core groupoid of $\Delta_{\perp}(C)$ is the discrete category 1 with one object. Clearly $R\Delta_{\perp}(C)$ is not Morita equivalent to $R1$.
- ▶ Let $\Delta'_{\perp}(C)$ denote the five-element semigroup obtained by omitting ι .
- ▶ **Important point for later:** $R\Delta'_{\perp}(C)$ (with multiplication coming from the semigroup) is actually an R -algebra **with identity element**

$$\ell_C = \sigma_0 - v + \sigma_1 .$$

A baby example with details, page 3

- In $R\Delta_{\perp}(C)$ we have a complete list of orthogonal idempotents:

$$\begin{aligned} e_0 &= \sigma_0 \sigma_1, \quad e_1 = (1 - \sigma_0) \sigma_1, \\ e_2 &= (1 - \sigma_1) \sigma_0, \quad e_3 = (1 - \sigma_1)(1 - \sigma_0), \end{aligned}$$

so that, in the “Cauchy completion” $\mathcal{Q}R\Delta_{\perp}(C)$ (obtained by freely splitting idempotents and freely adjoining direct sums), we have

$$C \cong E_0 \oplus E_1 \oplus E_2 \oplus E_3$$

where E_i is the object obtained in splitting the idempotent e_i .

- If we put $\gamma = \sigma_1 \sigma_0 - \sigma_0 \sigma_1$ and $\delta = \tau - \sigma_1$, using the relations in the presentation, we obtain the equations

$$e_1 \gamma = \gamma = \gamma e_2, \quad \delta e_1 = \delta = e_2 \delta, \quad \gamma \delta = e_1, \quad \delta \gamma = e_2$$

so $\gamma \vdash: E_1 \cong E_2$. It follows that we have an isomorphism of the form

$$C \cong D \oplus E \oplus E \oplus F.$$

A baby example with details, page 4

- ▶ Transporting the endomorphisms σ_0, σ_1, τ on C across this isomorphism yields the matrices

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

- ▶ Any endomorphism f of $D \oplus E \oplus E \oplus F$ which commutes with these three matrices must be of the form $f = f_0 \oplus f_1 \oplus f_1 \oplus f_2$.
- ▶ It follows that $R\Delta_{\perp}(C)$ is $\mathcal{V}$ -Morita equivalent to $R3$ where 3 is the discrete category with three objects. In other words, we have an equivalence of R -linear categories

$$[\Delta_{\perp}(C), \mathcal{V}] \cong \mathcal{V} \times \mathcal{V} \times \mathcal{V}.$$

Down to business: Factorization systems

1. Let $\mathcal{F}$ be a category with a factorization system $(\mathcal{E}, \mathcal{M})$. That is, $\mathcal{E}$ and $\mathcal{M}$ are sets of morphisms of $\mathcal{F}$ satisfying
 - FS0. for w invertible, $mw \in \mathcal{M}$ if $m \in \mathcal{M}$, while $we \in \mathcal{E}$ if $e \in \mathcal{E}$;
 - FS1. if $mh = ke$ with $e \in \mathcal{E}$ and $m \in \mathcal{M}$ then there exists a unique ℓ with $\ell e = h$ and $m\ell = k$;
 - FS2. every morphism f factors as $f = me$ for some $m \in \mathcal{M}$ and $e \in \mathcal{E}$.
2. It follows that $\mathcal{E} \cap \mathcal{M}$ is the set of invertible morphisms and that $\mathcal{E}$ and $\mathcal{M}$ are both closed under composition in $\mathcal{F}$.
3. We identify $\mathcal{E}$, $\mathcal{M}$ and $\mathcal{G} := \mathcal{E} \cap \mathcal{M}$ with the subcategories of $\mathcal{F}$ containing all objects.
4. Write $\mathcal{M}'$, $\mathcal{E}'$, $\mathcal{G}'$ for the sets of morphisms of $\mathcal{F}$ **not** in $\mathcal{M}$, $\mathcal{E}$, $\mathcal{G}$, respectively.

Proper factorization systems

The factorization system is called *proper* when

FSP. every member of $\mathcal{E}$ is an epimorphism and every member of $\mathcal{M}$ is a monomorphism.

If $(\mathcal{E}, \mathcal{M})$ is a proper factorization system on $\mathcal{F}$ then $(\mathcal{M}, \mathcal{E})$ is a proper factorization system on $\mathcal{F}^{\text{op}}$.

Proposition

In a proper factorization system, if $hf \in \mathcal{M}$ then $f \in \mathcal{M}$.

Dually, if $fk \in \mathcal{E}$ then $f \in \mathcal{E}$.

The bicategory $\mathfrak{R} := \mathcal{V}\text{-Mod}$

Objects of $\mathfrak{R}$ are $\mathcal{V}$ -categories (= R -linear categories or “ R -algebras with several objects”).

Hom categories are the $\mathcal{V}$ -functor $\mathcal{V}$ -categories

$$\mathfrak{R}(\mathcal{A}, \mathcal{B}) = [\mathcal{B}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}] ;$$

objects of these homs are called *modules from $\mathcal{A}$ to $\mathcal{B}$* .

These homcategories are all abelian R -linear categories so, within them, **exact sequences have meaning**.

Module composition

$$\mathfrak{R}(\mathcal{B}, \mathcal{C}) \otimes \mathfrak{R}(\mathcal{A}, \mathcal{B}) \xrightarrow{\circ} \mathfrak{R}(\mathcal{A}, \mathcal{C})$$

is defined by “ tensoring over $\mathcal{B}$ ”:

$$(K \circ H)(C, A) = \int^B H(B, A) \otimes K(C, B) .$$

More on $\mathfrak{R}$

- ▶ For $\mathcal{V}$ -functors $\mathcal{A} \xrightarrow{F} \mathcal{C} \xleftarrow{G} \mathcal{B}$, we have the module $\mathcal{C}(G, F) : \mathcal{A} \rightarrow \mathcal{B}$ with components $\mathcal{C}(G, F)(B, A) = \mathcal{C}(GB, FA)$.
- ▶ In particular, for each $\mathcal{V}$ -functor $F : \mathcal{A} \rightarrow \mathcal{B}$ we have the module $F_* = \mathcal{B}(1_{\mathcal{B}}, F) : \mathcal{A} \rightarrow \mathcal{B}$.
- ▶ Also, the module $F^* = \mathcal{B}(F, 1_{\mathcal{B}}) : \mathcal{B} \rightarrow \mathcal{A}$ provides a right adjoint $F_* \dashv F^*$ for F_* in $\mathfrak{R}$.
- ▶ If $\mathcal{B}$ is Cauchy complete then every module from $\mathcal{A}$ to $\mathcal{B}$ with a right adjoint in $\mathfrak{R}$ is isomorphic to F_* for some $\mathcal{V}$ -functor $F : \mathcal{A} \rightarrow \mathcal{B}$.
- ▶ $\mathcal{V}$ -categories are equivalent as objects of $\mathfrak{R}$ if and only if they are Morita equivalent (that is, they have equivalent presheaf $\mathcal{V}$ -categories).

Lemma on retracts of adjunctions

The lemma will be applied in $\mathfrak{R}$. It says that, if we have an idempotent 2-morphism on a morphism f with an adjoint u , then the splitting of that idempotent on f is a morphism with an adjoint obtained by splitting the mate idempotent on u . The form of the counit and unit is important. (A string proof is quite appealing.)

Lemma

In any bicategory $\mathfrak{M}$, suppose $f \dashv u : A \rightarrow X$ is an adjunction with counit $\varepsilon : fu \Rightarrow 1_A$ and unit $\eta : 1_X \Rightarrow uf$. Suppose $\omega : f \Rightarrow f$ is an idempotent 2-morphism on f with splitting provided by $\sigma : f \Rightarrow g$ and $\rho : g \Rightarrow f$: that is, $\rho\sigma = \omega$ and $\sigma\rho = 1_g$. The mate $\tilde{\omega} : u \Rightarrow u$ of $\omega : g \Rightarrow g$ under $f \dashv u$ is an idempotent 2-morphism on g satisfying $\varepsilon(\omega u) = \varepsilon(f\tilde{\omega})$. Then, any splitting of $\tilde{\omega}$ delivers a right adjoint v for g . Explicitly, if $\tilde{\rho}\tilde{\sigma} = \tilde{\omega}$ and $\tilde{\sigma}\tilde{\rho} = 1_v$ then $\tilde{\varepsilon} = \varepsilon(\rho\tilde{\rho})$ and $\tilde{\eta} = (\tilde{\sigma}\sigma)\eta$ provide a counit and unit for $g \dashv v$.

Modules as kernels of functorial transformations

- ▶ Every module $H : \mathcal{A} \rightarrow \mathcal{B}$ is the “kernel” of a functor $\widehat{H} : [\mathcal{B}, \mathcal{V}] \rightarrow [\mathcal{A}, \mathcal{V}]$ defined by composition with H in $\mathfrak{R}$ thus:
 $\widehat{H}(T) = (\mathcal{A} \xrightarrow{H} \mathcal{B} \xrightarrow{T} R\mathbf{1})$. That is,

$$\widehat{H}(T)A = \int^B H(B, A) \otimes TB . \quad (1)$$

- ▶ We obtain a pseudofunctor $\widehat{(-)} : \mathfrak{R}^{\text{op}} \rightarrow \mathcal{V}\text{-Cat}$ taking $\mathcal{A}$ to $[\mathcal{A}, \mathcal{V}]$ and H to $\widehat{H}$.
- ▶ Moreover, $\widehat{H}$ has a right adjoint $\widetilde{H} : [\mathcal{A}, \mathcal{V}] \rightarrow [\mathcal{B}, \mathcal{V}]$ given by right extension along H in $\mathfrak{R}$. Explicitly,

$$\widetilde{H}(F)B = \int_A [H(B, A), FA] = [\mathcal{A}, \mathcal{V}](H(B, -), F) . \quad (2)$$

The inclusion $j : R\mathcal{G} \hookrightarrow R\mathcal{F}$ gives the right adjoint $\mathcal{V}$ -module $j^* : R\mathcal{F} \rightarrow R\mathcal{G}$ where $j^*(A, B) = R\mathcal{F}(jA, B)$.

Lemma

For $g \in \mathcal{G}(C, A)$ and $h \in \mathcal{F}(B, D)$, there is a commutative square

$$\begin{array}{ccc}
 R\mathcal{M}'(A, B) & \xrightarrow{\subseteq} & R\mathcal{F}(A, B) \\
 \downarrow & & \downarrow R\mathcal{F}(g, h) \\
 R\mathcal{M}'(C, D) & \xrightarrow{\subseteq} & R\mathcal{F}(C, D)
 \end{array} \tag{3}$$

so that $M'(A, B) = R\mathcal{M}'(A, B)$ defines a submodule M' of j^* . There is a short exact sequence

$$0 \longrightarrow M' \xrightarrow{\subseteq} j^* \xrightarrow{q} M \longrightarrow 0 \tag{4}$$

in $\mathfrak{R}(R\mathcal{F}, R\mathcal{G})$ where $M(A, B) = R\mathcal{M}(jA, B)$.

Dually, there is a short exact sequence

$$0 \longrightarrow E' \xrightarrow{\subseteq} j_* \xrightarrow{\tilde{q}} E \longrightarrow 0$$

in $\mathfrak{K}(R\mathcal{G}, R\mathcal{F})$ where $E(B, A) = R\mathcal{C}(B, jA)$.

So now we have modules in both directions

$$M : R\mathcal{F} \rightarrow R\mathcal{G} \quad \text{and} \quad E : R\mathcal{G} \rightarrow R\mathcal{F} .$$

Do they give an equivalence in $\mathfrak{K}$?

Not in general, however:

$$M \circ E \xrightarrow{\cong} 1_{R\mathcal{G}}$$

Lemma

The family of R -module morphisms

$$\phi_B : E(B, D) \otimes M(C, B) \rightarrow R\mathcal{G}(C, D) ,$$

defined by, for $e \in \mathcal{E}(B, D)$ and $m \in \mathcal{M}(C, B)$,

$$\phi_B(e \otimes m) = \begin{cases} em & \text{for } em \in \mathcal{G} \\ 0 & \text{otherwise} \end{cases}, \text{ is dinatural in } B \in \mathcal{F}, \text{ natural in}$$

$C, D \in \mathcal{G}$, and induces an isomorphism

$$\bar{\phi} : M \circ E \xrightarrow{\cong} 1_{R\mathcal{G}}$$

in $\mathfrak{K}$ which gives an isomorphism $\hat{E} \circ \hat{M} \cong 1_{[\mathcal{G}, \mathcal{V}]}$. The mate of this isomorphism, under the adjunction $\hat{E} \dashv \tilde{E}$, yields a natural transformation

$$\Theta : \hat{M} \rightarrow \tilde{E} .$$

There are alternative formulas for $\hat{M}$ and $\tilde{E}$ which can be useful. For $A \in \mathcal{F}$, let $\text{Sub}A$ and $\text{Quo}A$ denote representative sets of the isomorphism classes of $\mathcal{M}/A$ and $A/\mathcal{C}$, respectively. Then

$$\hat{M}(T)A \cong \sum_{(U \xrightarrow{m} A) \in \text{Sub}A} TU \quad \text{and} \quad \tilde{E}(T)A \cong \prod_{(A \xrightarrow{e} W) \in \text{Quo}A} TW .$$

In cases where $\text{Sub}A$ and $\text{Quo}A$ are finite, both the sum and product in these alternative formulas are direct sums and the components of $\Theta : \hat{M} \rightarrow \tilde{E}$ transport across the isomorphisms to matrices whose non-zero entries have the form $T(em) : TU \rightarrow TW$ for $em \in \mathcal{G}$. Suppose that elements of each $\text{Sub}A$ can be listed $m_0, \dots, m_s$ and those of each $\text{Quo}A$ can be listed $e_0, \dots, e_s$, in such a way that $e_i m_j \in \mathcal{G}$ implies $1 \leq i \leq j \leq s$ and that $e_i m_i = 1$ for $0 \leq i \leq s$. Then these matrices are square and triangular with identities down the main diagonal, and so (because we have additive inverses in the hom R -modules) are invertible.

Example

Let $\mathcal{A}$ be any category in which every morphism is a monomorphism, pullbacks exist, and each slice category $\mathcal{A}/A$ has finitely many isomorphism classes. Take $\mathcal{F} = \mathcal{A}\sharp$ to be the category of spans in $\mathcal{A}$; that is, the objects are those of $\mathcal{A}$ and the morphisms $f : A \rightarrow B$ are isomorphism classes $[f_1, U, f_2]$ of spans $A \xleftarrow{f_1} U \xrightarrow{f_2} B$ in $\mathcal{A}$, where composition uses pullback. Take $\mathcal{M}$ to consist of those spans f with f_1 invertible and $\mathcal{E}$ to consist of those spans f with f_2 invertible. The core groupoid $\mathcal{G}$ of $\mathcal{F}$ is isomorphic to the core groupoid of $\mathcal{A}$. Here $\hat{M}$ is an equivalence; the Lack-St (2015) proof uses a comonadicity argument. An example of this is $\mathcal{A} = \mathbf{FI}$, the category of finite sets and injective functions; so $\mathcal{F} = \mathbf{FI}\sharp$. Then $\mathcal{G} = \mathfrak{S}$, the groupoid of finite sets and bijections, called *the symmetric groupoid*: so $[\mathcal{G}, \mathcal{V}]$ is the category of R -linear Joyal species; it is equivalent to the product over $n \geq 0$ of the categories of R -linear representations of the groups $\mathfrak{S}_n$. In this case, the deduced equivalence $[\mathbf{FI}\sharp, \mathcal{V}] \simeq [\mathfrak{S}, \mathcal{V}]$ is in Teimuraz Pirashvili (2000).

Plan to apply lemma on retracts of adjunctions

We continue with a proper factorization system $(\mathcal{E}, \mathcal{M})$ on a category $\mathcal{F}$. The goal is to find other conditions under which $M : R\mathcal{F} \rightarrow R\mathcal{G}$ in $\mathfrak{A}$ is an equivalence. If this is to be the case, M will need to have a left adjoint so we would like to apply our Lemma to deduce the adjoint from the adjunction $j_* \dashv j^*$. This means we would like M to be a retract of j^* . A natural splitting of the short exact sequence (4) would suffice. A splitting $\rho : M \rightarrow j^*$ of the epimorphism $q : j^* \rightarrow M$ has components

$$\rho_{A,B} : M(A, B) \rightarrow R\mathcal{F}(A, B)$$

which are natural in $A \in \mathcal{G}$ and $B \in \mathcal{F}$. In particular they are natural in $B \in R\mathcal{M}$ and so, by Yoneda, are determined by the value, say $\rho_A \in R\mathcal{F}(A, A)$, at $1_A \in R\mathcal{M}(A, A)$. Indeed $\rho_{A,B}(m) = m\rho_A$. This motivates the following assumption.

Idempotent Axiom. For each object $A \in \mathcal{F}$, there is a morphism $p_A : A \rightarrow A$ in $R\mathcal{F}$ such that

- p0. $p_A p_A = p_A$;
- p1. if $f \in \mathcal{F}(A, B)$ and $f \in \mathcal{M}'$ then $f p_A = 0$;
- p2. if $f \in \mathcal{F}(A, B)$ and $f \in \mathcal{E}'$ then $p_B f = 0$;
- p3. if $g \in \mathcal{G}(A, B)$ then $g p_A = p_B g$;
- p4. $p'_A = 1_A - p_A \in R(\mathcal{E}'(A, A) \cap \mathcal{M}'(A, A))$.

Example

Every groupoid $\mathcal{G}$ provides a trivial example with $\mathcal{F} = \mathcal{E} = \mathcal{M} = \mathcal{G}$ and $p_A = 1_A$. In the general case, we can think of $p'_A = 1_A - p_A$ as an obstruction to $\mathcal{F}$'s being a groupoid although we will see below the sense in which $\mathcal{F}$ is not too far from $\mathcal{G}$ at the R -linear level.

The retraction σ

Using p1, we have the identity linear function as the left side of a commutative square (5). Since the factorization system is proper, if $u \in \mathcal{M}'(A, A)$ and $f \in \mathcal{F}(A, B)$ then $fu \in \mathcal{M}'$; so, by p4, $fp'_A \in R\mathcal{M}'(A, B)$ for all $f \in \mathcal{F}(A, B)$. This gives a splitting $\sigma_{A,B}$ of the idempotent $R\mathcal{F}(p'_A, 1_B)$.

$$\begin{array}{ccc}
 R\mathcal{M}'(A, B) & \xrightarrow{\subseteq} & R\mathcal{F}(A, B) \\
 \downarrow 1 & \swarrow \sigma_{A,B} & \downarrow R\mathcal{F}(p'_A, 1_B) \\
 R\mathcal{M}'(A, B) & \xrightarrow{\subseteq} & R\mathcal{F}(A, B)
 \end{array} \tag{5}$$

Using p3, we see that the linear functions $R\mathcal{F}(p'_A, 1_B)$ are the components of an idempotent module morphism $\pi' : j^* \rightarrow j^*$ which is split by the module morphism $\sigma : j^* \rightarrow M'$ and the inclusion $M' \hookrightarrow j^*$. It follows that the short exact sequence for M splits so that, if we put $\pi = 1 - \pi' = R\mathcal{F}(p_A, 1_B)$, we have:

Lemma

The idempotent module morphism $\pi = R\mathcal{F}(p_A, 1_B) : j^* \rightarrow j^*$ splits as

$$\begin{array}{ccc}
 j^* & \xrightarrow{\pi} & j^* \\
 \searrow q & & \nearrow \rho \\
 & M & \xrightarrow{1_M} M \\
 & & \searrow q
 \end{array}
 \quad (6)$$

where, for $m \in \mathcal{M}(A, B)$, $\rho_{A,B}(m) = mp_A$ and for $g \in \mathcal{G}(A', A)$ and $f \in \mathcal{F}(B, B')$,

$$M(g, f)m = fmgp_{A'}.$$

Dually, we also have:

Lemma

The idempotent module morphism $\varpi = R\mathcal{F}(1_A, p_B) : j_* \rightarrow j_*$ splits as

$$\begin{array}{ccccc}
 j_* & \xrightarrow{\varpi} & j_* & & \\
 & \searrow \tilde{q} & \nearrow \tilde{\rho} & \searrow \tilde{q} & \\
 & & E & \xrightarrow{1_E} & E
 \end{array} \tag{7}$$

where $E(A, B) = R\mathcal{E}(A, B)$, $\tilde{\rho}_{A,B}(e) = p_B e$, and

$$E(f, g)e = p_{B'} g e f ,$$

for $e \in \mathcal{E}(A, B)$, $f \in \mathcal{F}(A', A)$, $g \in \mathcal{E}(B, B')$.

Mates splitting

Lemma

The idempotent module morphisms $\pi : j^ \rightarrow j^*$ and $\varpi : j_* \rightarrow j_*$ are mates under the adjunction $j_* \dashv j^*$ in $\mathfrak{K}$. By the retract-of-adjunction Lemma, $E \dashv M$ in $\mathfrak{K}$.*

Since $\widehat{(-)} : \mathfrak{K}^{\text{op}} \rightarrow \mathcal{V}\text{-Cat}$ and $\widetilde{(-)} : \mathfrak{K}^{\text{co}} \rightarrow \mathcal{V}\text{-Cat}$ are pseudofunctors:

Corollary

The functor $\widehat{M} : [\mathcal{G}, \mathcal{V}] \rightarrow [\mathcal{F}, \mathcal{V}]$ has right adjoint $\widehat{E}$. The functor $\widetilde{M} : [\mathcal{F}, \mathcal{V}] \rightarrow [\mathcal{G}, \mathcal{V}]$ has right adjoint $\widetilde{E}$. So $\widehat{E} \cong \widetilde{M}$.

Lemma

The unit $1_{R\mathcal{G}} \Rightarrow M \circ E$ of the adjunction $E \dashv M$ in $\mathfrak{X}$ is invertible. So the unit $1_{[\mathcal{G}, \mathcal{V}]} \Rightarrow \hat{E} \circ \hat{M}$ of $\hat{M} \dashv \hat{E}$ is invertible and $\hat{M}$ is fully faithful.

Proof.

Note that $(j^* \circ j_*)(C, D) \cong R\mathcal{F}(jC, jD)$ and, from the adjunction retract lemma, the unit of the adjunction $E \dashv M$ has component at (C, D) equal to the composite

$$R\mathcal{G}(C, D) \xrightarrow{j} R\mathcal{F}(jC, jD) \cong \int^{B \in \mathcal{F}} R\mathcal{F}(B, jD) \otimes R\mathcal{F}(jC, B) \\ \xrightarrow{\int^{B \in \mathcal{F}} q \otimes \tilde{q}} (M \circ E)(C, D)$$

which is inverse to the component of our earlier isomorphism $\bar{\phi}$. □

The hardest bit

Lemma

If each object A of $\mathcal{F}$ has only finitely many $\mathcal{M}$ -subobjects (that is, each slice category $\mathcal{M}/A$ has only finitely many isomorphism classes) then the counit of the adjunction $E \dashv M$ in $\mathfrak{R}$ is a split epimorphism.

Proof.

The counit, which is natural, is the composite

$$\begin{aligned} (E \circ M)(A, B) &= \int^{C \in \mathcal{G}} R\mathcal{M}(C, B) \otimes R\mathcal{E}(A, C) \\ &\rightarrow (j_* \circ j^*)(A, B) \rightarrow R\mathcal{F}(A, B) \end{aligned}$$

which takes the equivalence class of $m \otimes e \in R\mathcal{M}(C, B) \otimes R\mathcal{E}(A, C)$ to $mp_C e \in R\mathcal{F}(A, B)$. We claim this family has a natural right inverse.

To be continued.



Proof continued.

In particular, for $A = B$, we must see that the identity 1_A of A is in the image. Take a finite family $(C_i \xrightarrow{m_i} A)_{i=0}^k$ of morphisms in $\mathcal{M}$ representing all isomorphism classes in the ordered set $\mathcal{M}/A$ and with the property that $m_i = m_j n$ for some $C_i \xrightarrow{n} C_j$ implies $i \leq j$; we can suppose $C_k = A$ and $m_k = 1_A$. Let

$$\phi_j : \bigoplus_{i \leq j} R\mathcal{M}(C_i, A) \otimes R\mathcal{E}(A, C_i) \longrightarrow R\mathcal{F}(A, A)$$

denote the function defined by $\phi_j(m \otimes e) = mp_{C_i}e$ for $m \in R\mathcal{M}(C_i, A)$ and $e \in R\mathcal{E}(A, C_i)$. So the image of ϕ_i is contained in the image of ϕ_j for $i \leq j$. By induction on j one can show that every $f \in \mathcal{F}(A, A)$ is in the image of the function ϕ_k .

Hence, $1_A \in \mathcal{F}(A, A)$ is in the image of the A, A component of the counit.

To be continued.



Proof continued further.

Suppose $t \in \int^{C \in \mathcal{G}} R\mathcal{M}(C, A) \otimes R\mathcal{E}(A, C)$ maps to $1_A \in \mathcal{F}(A, A)$. By the Yoneda Lemma, there exists a unique family of morphisms

$$R\mathcal{F}(A, B) \longrightarrow \int^{C \in \mathcal{G}} R\mathcal{M}(C, B) \otimes R\mathcal{E}(A, C)$$

which is natural in $B \in \mathcal{F}$ taking $1_A \in \mathcal{F}(A, A)$ to t . Again by Yoneda, this gives right inverses to the components of the counit. Dually, there is such a family natural in B . By Yoneda yet again, these families agree since they have the same value at the identity. So the counit of $E \dashv M$ has a natural right inverse.

That's really the end of proof!



Main Theorem

Theorem

If each object A of $\mathcal{F}$ has only finitely many $\mathcal{M}$ -subobjects then the adjunction $E \dashv M$ is an equivalence $R\mathcal{F} \simeq R\mathcal{G}$ in $\mathfrak{R}$.

Proof.

We now know that the unit $1_{R\mathcal{G}} \Rightarrow \tilde{M} \circ \hat{M}$ is invertible. So $\hat{M} : [R\mathcal{G}, \mathcal{V}] \rightarrow [R\mathcal{F}, \mathcal{V}]$ is fully faithful and so is its composite $M' : R\mathcal{G}^{\text{op}} \rightarrow [R\mathcal{F}, \mathcal{V}]$, $C \mapsto M(C, -)$, with the Yoneda embedding. Moreover, one of our Lemmas implies M' is strongly generating. By Theorem 2 of Day-St (1986, about strong generators being dense), M' is also dense. So the counit of $E \dashv M$ is invertible. □

In the terminology of ring theory, this Theorem implies that $R\mathcal{F}$ and $R\mathcal{G}$ are Morita equivalent several-object R -algebras. In the terminology of enriched category theory, it implies that $R\mathcal{F}$ and $R\mathcal{G}$ are Cauchy equivalent $\mathcal{V}$ -categories.

Consequences

Corollary

*The functor $\widehat{M} : [\mathcal{G}, \mathcal{V}] \rightarrow [\mathcal{F}, \mathcal{V}]$ is an equivalence with pseudo-inverse $\widehat{E}$.
The functor $\widetilde{M} : [\mathcal{F}, \mathcal{V}] \rightarrow [\mathcal{G}, \mathcal{V}]$ is an equivalence with pseudo-inverse $\widetilde{E}$.*

Corollary

The equivalence $\widetilde{M}$ is a retract of the restriction functor

$$[j, 1] : [R\mathcal{F}, \mathcal{V}] \rightarrow [R\mathcal{G}, \mathcal{V}] .$$

Corollary

For any R -linear category $\mathcal{X}$ in which idempotents split and finite direct sums exist, there is an equivalence

$$[\mathcal{F}, \mathcal{X}] \simeq [\mathcal{G}, \mathcal{X}] .$$

Stiffness

Definition

In any category $\mathcal{F}$, a morphism $f : A \rightarrow B$ is called *stiff* when the only endomorphisms of A through which f factors are automorphisms. In other words, $f = (A \xrightarrow{u} A \xrightarrow{v} B)$ implies u invertible. A morphism is *costiff* when it is stiff in the opposite category. A category is called *stiff* when the costiff and stiff morphisms are the $\mathcal{E}$ and $\mathcal{M}$ of a proper factorization system.

The category of finite sets and the category of finite-dimensional vector spaces over a field are stiff categories: costiff = surjective, stiff = injective. Any stiff endomorphism is an automorphism so a consequence of stiffness is the **pigeon-hole principle**:

Proposition (PhP)

For all objects A in a stiff category $\mathcal{F}$, the inclusions $\mathcal{M}(A, A) \subseteq \mathcal{G}(A, A)$ and $\mathcal{E}(A, A) \subseteq \mathcal{G}(A, A)$ are equalities. Equivalently, if $\mathcal{G}(A, B) \neq \emptyset$ then $\mathcal{G}(A, B) = \mathcal{E}(A, B) = \mathcal{M}(A, B)$.

Some work of Laci Kovács (1992)

following W. Munn (1957) and I.S. Ponizovskii (1956)

Recall that the set $\mathcal{G}'$ of morphisms is the complement of $\mathcal{G}$ in $\mathcal{F}$.

For each $C \in \mathcal{F}$, **PhP** implies that $R\mathcal{G}'(C, C)$ is a two-sided ideal in $R\mathcal{F}(C, C)$. In particular $R\mathcal{G}'(C, C)$ is an R -algebra, possibly without an identity element.

In the case where $\mathcal{F} = \text{vect}_{\mathbb{F}}$ is the category of finite vector spaces over a finite field $\mathbb{F}$ and the factorization is surjective-injective $\mathbb{F}$ -linear functions, Laci Kovács (1992) produced an identity element making $R\mathcal{G}'(C, C)$ a unital algebra **provided the characteristic of $\mathbb{F}$ is invertible in R** . Also see Subsection 3.1 of Kuhn's paper for helpful material on this.

Therefore we make the following definition in the general situation.

Definition

An *MPK idempotent* is an identity element ℓ_C making $R\mathcal{G}'(C, C)$ an R -algebra under composition.

From MPK to the Idempotent Axiom

Lemma

Assume *PhP* and that each object C has an MPK idempotent ℓ_C . Then ℓ_C is a central idempotent in the R -algebra $R\mathcal{F}(C, C)$ and the morphisms $\ell_C : C \rightarrow C$ are the components of a natural endomorphism of the inclusion functor of $j : R\mathcal{G} \rightarrow R\mathcal{F}$.

Proposition

Assume that $\mathcal{F}$ is stiff and that each object C has a MPK idempotent ℓ_C . Then the idempotents $p_A = 1_A - \ell_A : A \rightarrow A$ satisfy the Idempotent Axiom.

Semisimplicity

It can happen that $R\mathcal{F}(C)$ is semisimple. This is true when $Y = \mathcal{F}(C)$ is any finite monoid of Lie type and R is a field of characteristic 0 [Okniński-Putcha (1991)]. Steinberg's 2016 book provides necessary and sufficient conditions on a finite monoid Y and field R in order for RY to be semisimple; in particular, the characteristic of R should not divide the order of the group of invertible elements in the monoid pYp for any idempotent $p \in Y$. Itamar Stein has proved all the endomorphism monoids $R\Delta_{\perp}(\mathbf{a})$ of $R\Delta_{\perp}$ are semisimple for R any field.

With semisimplicity of RY , the inclusion $J \hookrightarrow RY$ of every two-sided ideal J of RY splits as a left module morphism and splits as a right module morphism. The value of the left module splitting at the identity of RY gives a right identity for J and the value of the right module splitting at the identity of RY gives a left identity for J . So J becomes an R -algebra with identity. This is a source of MPK idempotents for categories $\mathcal{F}$ with endomorphism monoids $\mathcal{F}(C)$ finite and $R\mathcal{F}(C)$ semisimple.

Thank you very much for the invitation to talk and for
your attention 😊

